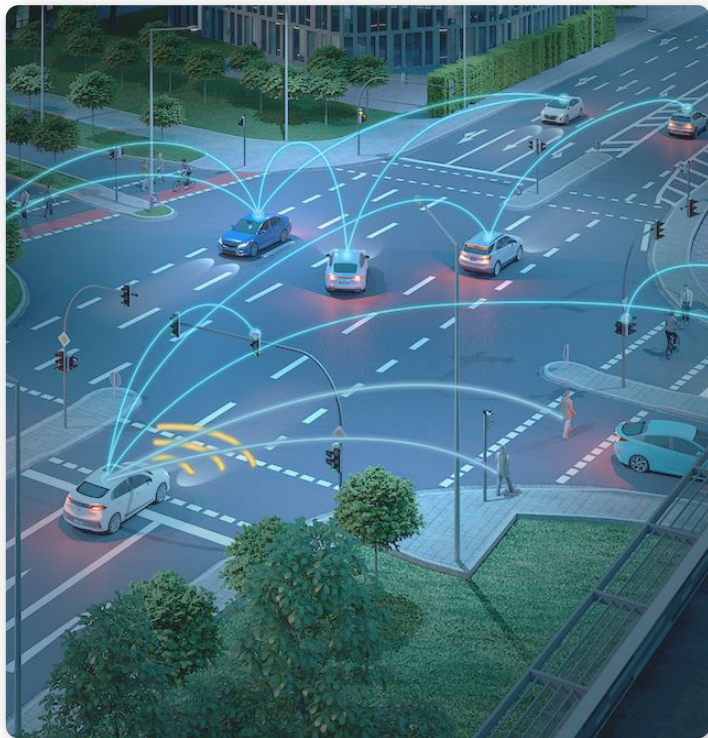
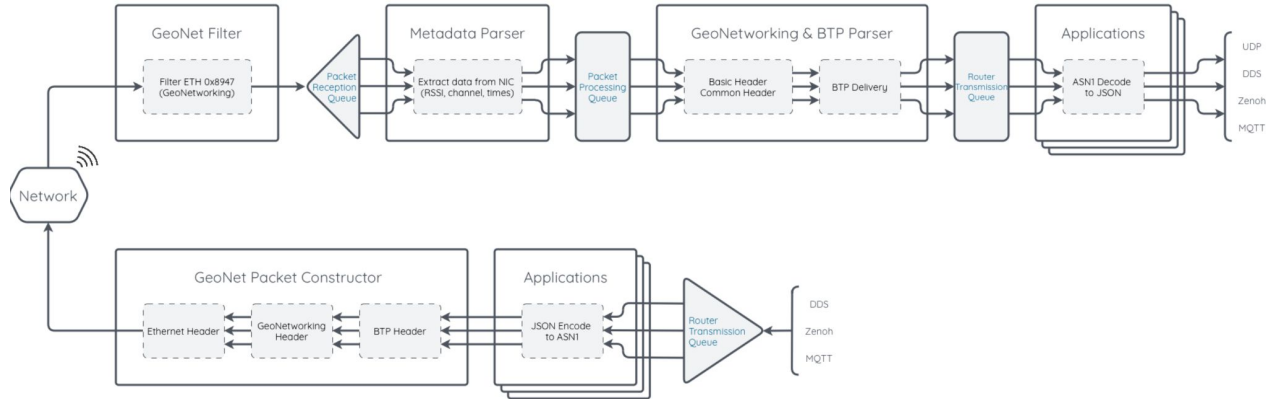




What is Vanetza-NAP?

Overview

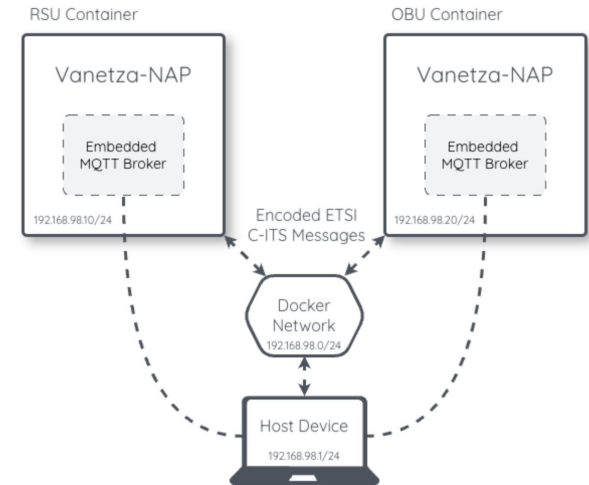
- Open source C++ implementation of the **ETSI C-ITS** protocol suite for **V2X communications**.
- Designed to handle **ETSI** messages in vehicular networks.
- Works over:
 - ◇ **ITS-G5 (802.11p)**.
 - ◇ Geonetworking over **IP multicast**.
- Encodes & Decodes **ITS** messages (CAM, DENM,...) using **ASN.1**.
- Handles GeoNetworking (location-based packet routing) and the full stack from radio to application.



Extension

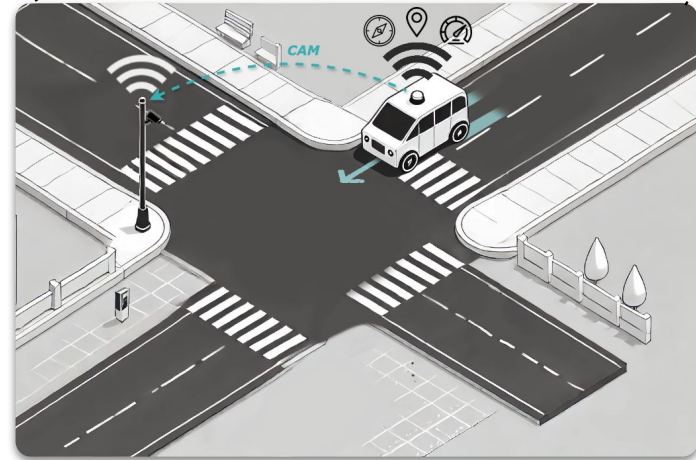
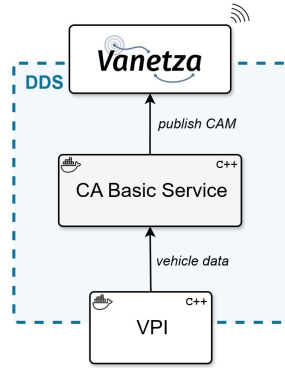
Vanetza-NAP

- Can be deployed on a single machine through docker and **simulate multiple ITS stations** without real hardware
- Send and receive ETSI messages as **simple JSON payloads** through MQTT / DDS / Zenoh
- Support for **additional** ETSI messages (MAPEM, SPATEM, MVM, MIM, ...)
- Possibility of exchanging messages through **UDP multicast groups**
- One single NAP-Vanetza can **emulate many different stations** using the same hardware (different *stationIDs*/MAC addresses)
- **Unit transformations** between human-readable and ETSI standards



Use Cases - CAMs

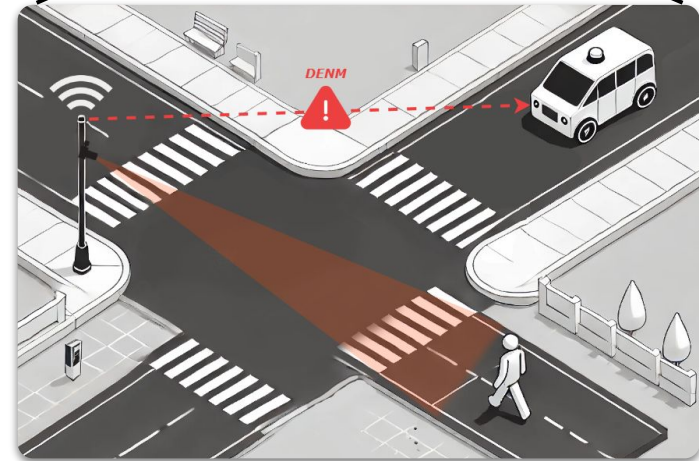
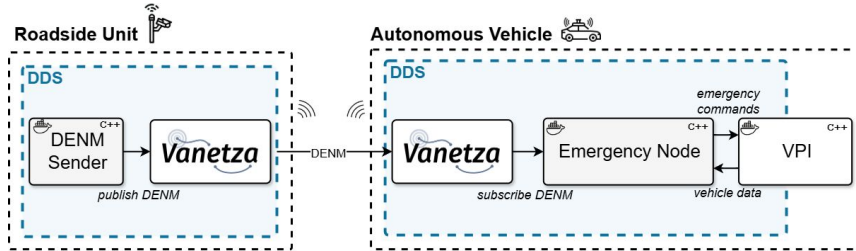
Share the station's real-time status to the surrounding V2X network through CAMs



Use Cases - DENMs

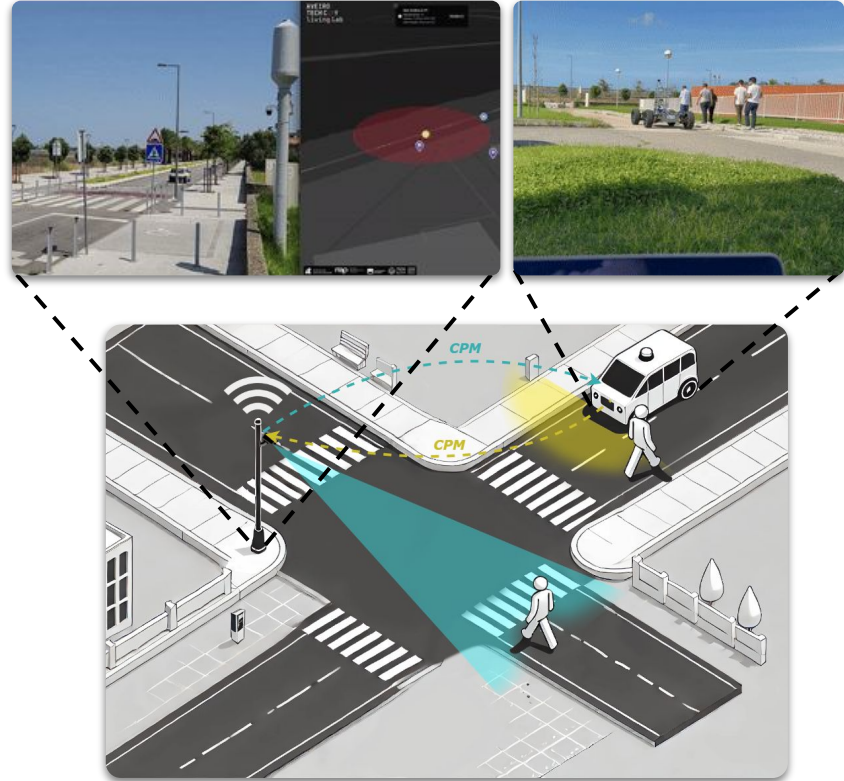
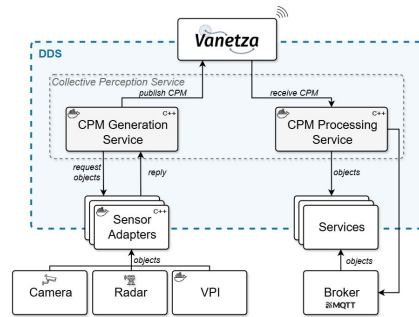
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Share and process DENMs to alter the vehicle's emergency behaviour



Use Cases - CPMs

Share information about road users, obstacles and sensing capabilities through CPMs



Use Cases - MAPEMs & SPATEMs

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Enhance vehicle navigation at intersections through SPATEMs and MAPEMs

